

Wisper Alpha

Automated Passive Reconnaissance & Asset Intelligence Platform

Project Vision

A local-first reconnaissance platform designed to collect, correlate, and score external attack-surface intelligence through an automated, session-isolated workflow.

Prepared for: CSAI 301/203 Technical Review

Focus: Passive Intelligence, Session Isolation, and Evidence Traceability

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Contents

1	Executive Summary	2
2	System Architecture	2
2.1	Architecture Visualization	2
2.2	Core Components	2
3	Operational Modes & Pipeline	3
3.1	Pipeline Lifecycle	3
4	Data and Security Planning	3
4.1	Session Isolation	3
4.2	Confidence Model	3
5	Security and Ethics	4
6	Conclusion	4

1 Executive Summary

Manual reconnaissance in modern cybersecurity is often fragmented, slow, and prone to human error. **Wisper Alpha** addresses these challenges by providing an automated pipeline that transforms raw external data into actionable asset intelligence. By utilizing a "Session-Aware" architecture, the platform ensures strict scope locking and data isolation for different target websites, allowing researchers to switch between projects without risk of data contamination.

2 System Architecture

The system follows a modular, adapter-based design to ensure scalability and ease of integration for new intelligence providers.

2.1 Architecture Visualization

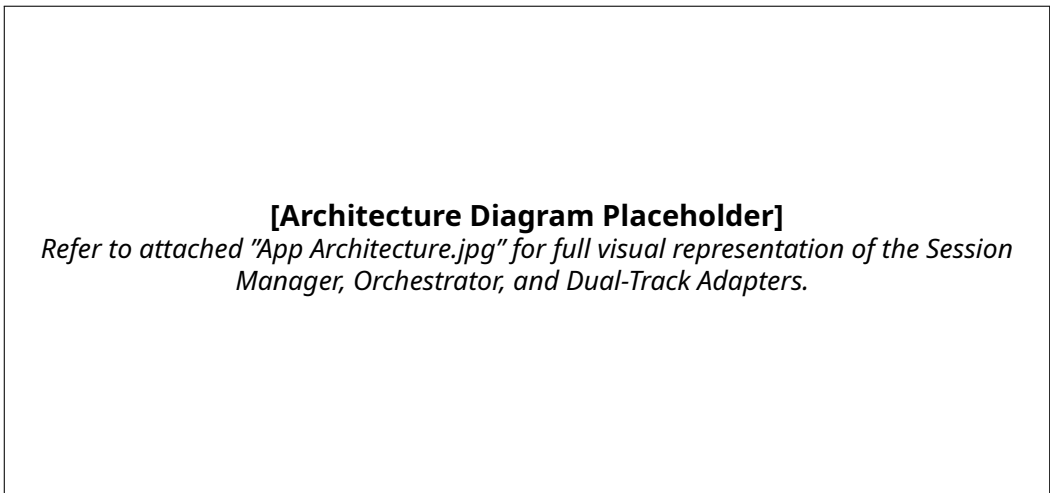


Figure 1: High-Level System Flow: From Web UI Intake to Advanced Delta Reporting.

2.2 Core Components

- Web UI:** A centralized console for target management, session switching, and real-time dashboard viewing.
- Backend API:** Manages auth, job queue submission, and enforces scope locking via the Session Manager.
- Session Manager:** The "gatekeeper" of the system. It ensures that every recon run is confined to a specific scope and profile (Core vs. Advanced).
- Orchestrator:** Handles the execution logic, rate limiting, and provider policies in a deterministic order.
- Adapter Layer:** Features a dual-track approach. **Core Adapters** handle baseline passive discovery, while **Advanced Adapters** perform deep enrichment (DNS/TLS/ASN Mapping).
- Correlation & Scoring:** Merges entities from disparate sources and assigns confidence scores based on source diversity and consistency.

3 Operational Modes & Pipeline

Wisper Alpha distinguishes between speed and depth through its two primary profiles.

Table 1: Operating Profile Comparison

Feature	Core Mode (Baseline)	Advanced Mode (Deep)
Primary Goal	Fast passive discovery	High-confidence intelligence
Discovery	Passive Subdomains (Amass, Subfinder)	Deep DNS, TLS, and ASN Mapping
Collection	Historical URLs (Wayback, GAU)	Controlled Crawling (Katana)
Runtime	Short (Minutes)	Long (Hours)
Usage	Initial assessment	Detailed target profiling

3.1 Pipeline Lifecycle

The pipeline follows a strict progression:

1. **Target Intake:** Definition of scope and session initialization.
2. **Passive Discovery:** Broad identification of subdomains and certificates.
3. **Profile Branching:** Execution of Core or Advanced enrichment tools.
4. **Normalization:** Conversion of multi-source outputs into a canonical schema.
5. **Scoring:** Applying the confidence model to prioritize findings.
6. **Reporting:** Generating executive summaries with "Advanced-Delta" highlighting.

4 Data and Security Planning

4.1 Session Isolation

To prevent data contamination, Wisper Alpha implements a strict hierarchy:

- **Scope Lock:** A session is bound to a specific domain or CIDR range.
- **Data Segregation:** Every record in the database is tagged with a *Session_ID*.
- **Comparison Logic:** Cross-session data viewing is disabled unless the "Explicit Compare" module is invoked.

4.2 Confidence Model

Confidence is not a static value but a dynamic score calculated as:

$$\text{Score} = \frac{\sum(\text{Source Diversity} \times \text{Source Reliability})}{\text{Contradiction Penalty} \times \text{Age Factor}}$$

5 Security and Ethics

Wisper Alpha is built on a **Passive-First** policy.

- **Authorization:** Users must confirm authorization before initializing a session.
- **Rate Limiting:** The Orchestrator enforces provider-specific delays to prevent DoS-like behavior against third-party APIs.
- **Traceability:** Every finding is linked back to its raw evidence record to ensure transparency in reporting.

6 Conclusion

Wisper Alpha provides a robust, course-grade framework for modern reconnaissance. By emphasizing session isolation and the distinction between Core and Advanced modes, it offers a scalable solution for researchers to maintain high-quality, reproducible intelligence while respecting technical and ethical boundaries.